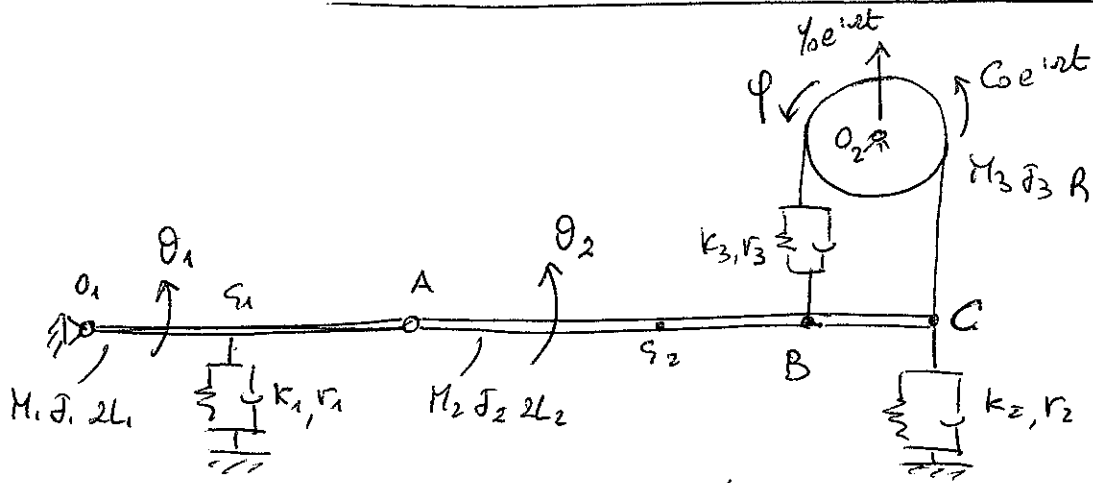
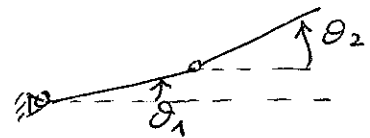




$$\underline{x}_L = \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} \rightarrow \underline{x} = \begin{Bmatrix} \underline{x}_L \\ \underline{x}_v \end{Bmatrix} = \begin{Bmatrix} \theta_1 \\ \theta_2 \\ y \end{Bmatrix}$$

$\theta_2 = \text{rotazione assoluta}$



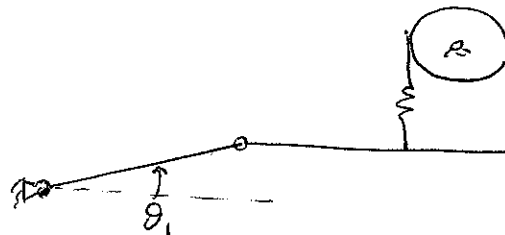
$$E_c = \frac{1}{2} M_1 \dot{y}_{q1}^2 + \frac{1}{2} J_1 \dot{\theta}_1^2 + \frac{1}{2} M_2 \dot{y}_{q2}^2 + \frac{1}{2} J_2 \dot{\theta}_2^2 + \frac{1}{2} M_3 \dot{y}^2 + \frac{1}{2} J_3 \dot{\varphi}^2$$

$$[m_F] = \text{diag}(M_1, J_1, M_2, J_2, M_3, J_3); \quad \underline{y}_{mF} = \{y_{q1}, \theta_1, y_{q2}, \theta_2, y, \varphi\}^T$$

$$E_c = \frac{1}{2} \dot{\underline{y}}_{mF}^T [m_F] \dot{\underline{y}}_{mF}$$

lezioni cinematiche:

• $\theta_1 \neq 0 \quad \theta_2 = \varphi = 0$



$$y_{q1} = L_1 \theta_1$$

$$y_{q2} = 2L_1 \theta_1$$

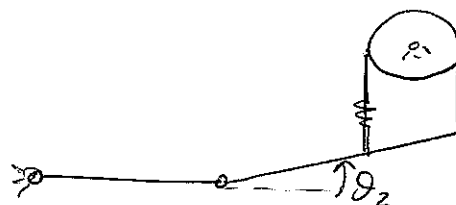
$$y_c = y_{q2} \rightarrow \varphi = \frac{2L_1 \theta_1}{R}$$

$$\Delta L_1 = L_1 \theta_1$$

$$\Delta L_2 = 2L_1 \theta_1$$

$$\Delta L_3 = -y_c - y_B = -4L_1 \theta_1$$

• $\theta_1 = 0 \quad \theta_2 \neq 0 \quad \varphi = 0$



$$y_{q1} = 0$$

$$y_{q2} = L_2 \theta_2$$

$$y_c = 2L_2 \theta_2 \rightarrow \varphi = \frac{2L_2 \theta_2}{R}$$

$$\Delta L_1 = 0$$

$$\Delta L_2 = 2L_2 \theta_2$$

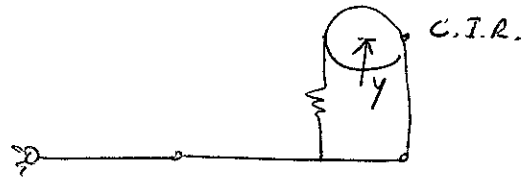
$$\Delta L_3 = -y_c - y_B = -2L_2 \theta_2 - (2L_2 - 2R) \theta_2 = (-4L_2 + 2R) \theta_2$$

$$\theta_1 = \theta_2 = 0 \quad y \neq 0$$

$$y_{q1} = y_{q2} = \Delta L_1 = \Delta L_2 = 0$$

$$\varphi = -y/R$$

$$\Delta L_3 = 2y$$



$$[\Lambda_{uu}] = \begin{bmatrix} L_1 & 0 & 0 \\ 1 & 0 & 0 \\ 2L_1 & L_2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{2L_1}{R} & \frac{2L_2}{R} & -\frac{1}{R} \end{bmatrix}$$

$$(\Lambda_{Ee}) = \begin{bmatrix} L_1 & 0 & 0 \\ 2L_1 & 2L_2 & 0 \\ -4L_1 & -4L_2 & 2 \end{bmatrix}$$

$$V_E = \frac{1}{2} k_1 \Delta L_1^2 + \frac{1}{2} k_2 \Delta L_2^2 + \frac{1}{2} k_3 \Delta L_3^2$$

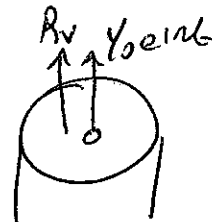
$$V_q = m_1 g h_{q1} + m_2 g h_{q2} + m_3 g h_{q3}$$

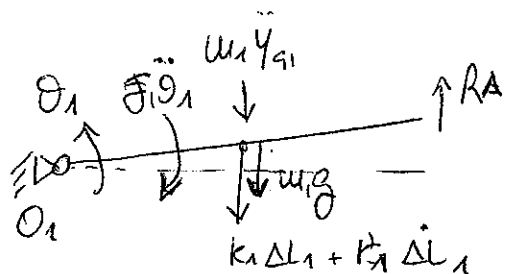
$$\left. \begin{aligned} h_{q1} &= L_1 \sin \theta_1 \\ h_{q2} &= 2L_1 \sin \theta_1 + L_2 \sin \theta_2 \\ h_{q3} &= y \rightarrow \text{LINEARE} \end{aligned} \right\} \rightarrow \frac{\partial^2}{\partial x^2} = 0 \rightarrow [k_q] = \phi$$

$$\delta L = C_0 e^{i\omega t} \delta \varphi + R_v \delta y =$$

$$= C_0 e^{i\omega t} \left(\frac{2L_1}{R} \delta \theta_1 + \frac{2L_2}{R} \delta \theta_2 - \frac{\delta y}{R} \right) + R_v \delta y$$

$$\rightarrow \underline{Q}_L = \begin{bmatrix} 2L_1/R \\ 2L_2/R \end{bmatrix} C_0 e^{i\omega t}$$





$$\sum M_{O_1} = 0 : \quad R_A 2L_1 - (w_1 \ddot{y}_{G1} + k_1 \Delta L_1 + r_1 \dot{\Delta L}_1) L_1 + J_1 \ddot{\theta}_1 - w_1 g L_1 = 0$$

$$R_A = \frac{w_1 \ddot{y}_{G1} + k_1 \Delta L_1 + r_1 \dot{\Delta L}_1 + w_1 g}{2} + \frac{J_1 \ddot{\theta}_1}{2L_1}$$

$$\ddot{\theta}_1 = -\Omega^2 \theta_1$$

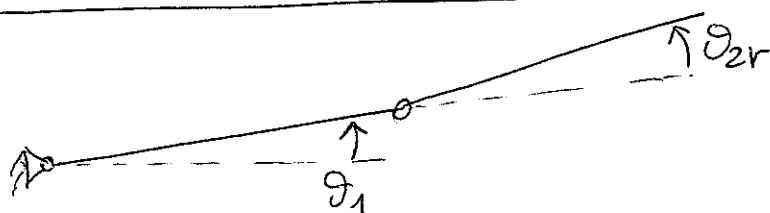
$$\ddot{y}_{G1} = -\Omega^2 \theta_1 L_1$$

$$\Delta L_1 = L_1 \theta_1$$

$$\dot{\Delta L}_1 = \Omega L_1 \theta_1$$

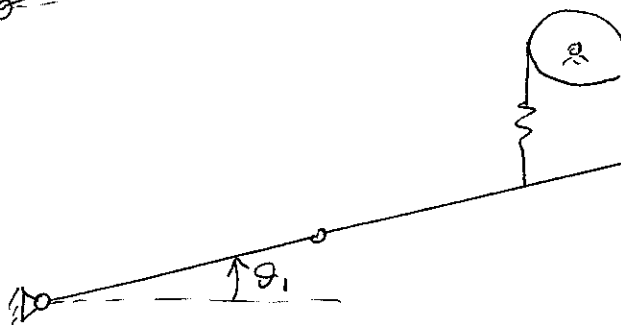
NB: non sono stati considerati i termini di ordine superiore al 1. $\cos \theta_1 \approx 1$

SOLUZIONE CON θ_1 e θ_{2r} rel



legami cinematici:

$$\bullet \theta_1 \neq 0 \quad \theta_{2r} = \varphi = 0$$



$$y_{G1} = L_1 \theta_1$$

$$y_{G2} = (2L_1 + L_2) \theta_1$$

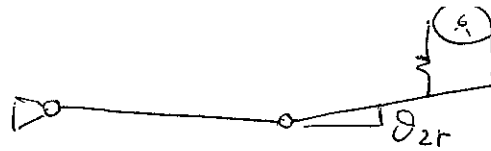
$$y_C = (2L_1 + 2L_2) \theta_1 \rightarrow \varphi = \frac{(2L_1 + 2L_2) \theta_1}{R}$$

$$\Delta L_1 = L_1 \theta_1$$

$$\Delta L_2 = (2L_1 + 2L_2) \theta_1$$

$$\Delta L_3 = -y_C - y_B = -(4L_1 + 4L_2 - 2R) \theta_1$$

$$\cdot \vartheta_1 = 0 \quad \vartheta_{2r} \neq 0 \quad \gamma = 0$$



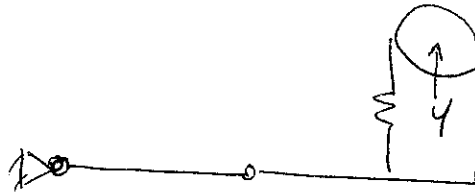
$$\gamma_{s1} = \Delta L_1 = 0$$

$$\gamma_{s2} = L_2 \vartheta_{2r}$$

$$\gamma_c = \Delta L_2 = 2L_2 \vartheta_{2r} \rightarrow \psi = \frac{2L_2 \vartheta_{2r}}{R}$$

$$\Delta L_3 = (-4L_2 + 2R) \vartheta_{2r}$$

$$\cdot \vartheta_1 = \vartheta_{2r} = 0 \quad \gamma = 0$$



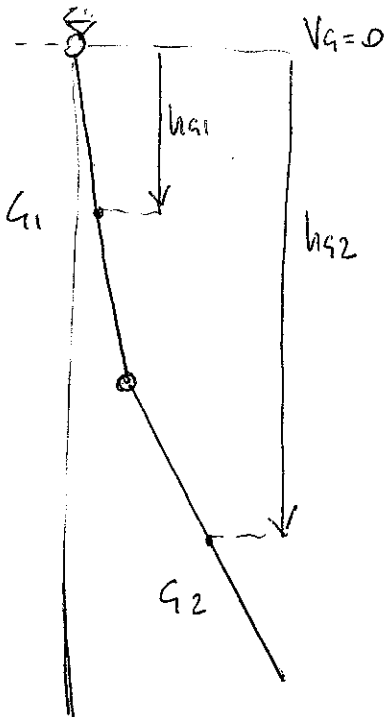
$$\gamma_{s1} = \gamma_{s2} = \Delta L_1 = \Delta L_2 = 0$$

$$\psi = -\gamma/R$$

$$\Delta L_3 = 2\gamma$$

SOLUZIONE DEL TESTO B

- per i legami cinematici si veda la soluzione del Testo A
- Energia potenziale gravitazionale:



$$V_g = m_1 g h_{s1} + m_2 g h_{s2}$$

$$\cdot \underline{x} = \begin{Bmatrix} \vartheta_1 \\ \vartheta_2 \end{Bmatrix} \rightarrow \begin{aligned} h_{s1} &= -L_1 \cos \vartheta_1 \\ h_{s2} &= -2L_1 \cos \vartheta_1 - L_2 \cos \vartheta_2 \end{aligned}$$

$$V_g = -(m_1 g L_1 + 2m_2 g L_1) \cos \vartheta_1 - m_2 g L_2 \cos \vartheta_2 \rightarrow$$

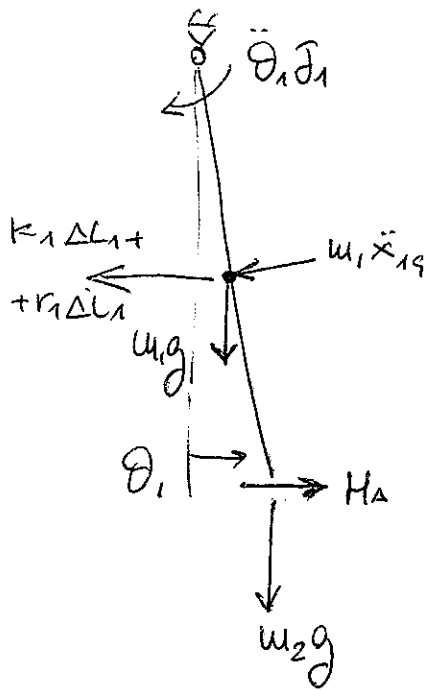
$$\rightarrow (K_g) = \begin{bmatrix} m_1 g L_1 + 2m_2 g L_1 & 0 & 0 \\ 0 & m_2 g L_2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\cdot \underline{x} = \begin{Bmatrix} \vartheta_1 \\ \vartheta_{2r} \end{Bmatrix} \rightarrow \begin{aligned} h_{s1} &= -L_1 \cos \vartheta_1 \\ h_{s2} &= -2L_1 \cos \vartheta_1 - L_2 \cos(\vartheta_1 + \vartheta_{2r}) \end{aligned}$$

$$V_g = -(m_1 g L_1 + 2m_2 g L_1) \cos \vartheta_1 - m_2 g L_2 \cos(\vartheta_1 + \vartheta_{2r})$$

$$\rightarrow [K_s] = \begin{bmatrix} m_1 g L_1 + 2m_2 g L_1 + m_2 g L_2 & m_2 g L_2 & 0 \\ m_2 g L_2 & m_2 g L_2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

- ressource orientierte in A



$$\sum M_{O_1} = 0 :$$

$$H_A 2L_1 \cos \vartheta_1 - m_1 L_1 \ddot{x}_{1c} - J_1 \ddot{\theta}_1 - m_2 g 2L_1 \sin \vartheta_1 +$$

$$- (\kappa_1 \Delta L_1 + r_1 \Delta \dot{L}_1) L_1 \cos \vartheta_1 - m_1 g L_1 \sin \vartheta_1 = 0$$

$$\begin{aligned}\cos \vartheta_1 &\approx 1 \\ \sin \vartheta_1 &\approx \vartheta_1\end{aligned}$$

$$H_A = \frac{m_1 \ddot{x}_{A1} + 2m_2 g \vartheta_1 + (k_1 \Delta L_1 + F_1 \Delta L_1)}{2} + \frac{m_1 g \vartheta_1}{2} + \frac{J_1 \ddot{\vartheta}_1}{2L_1}$$